

HW6 NEW SKlearn - Full AI Engineer ML Project

Complete sklearn workflow: preprocessing, regression, classification, PCA, clustering, benchmarking, joblib deployment, and Streamlit app.

HW6 NEW SKlearn - Full AI Engineer ML Project

Preprocessing + Regression + Classification + PCA + KMeans + Benchmark + Deployment

Data
1000 rows
Augmented regression dataset

Best regression
KNN Regressor
R2 0.989

Best classifier
Logistic Regressi...
F1 0.916

PCA
2D
Var 92.3%

KMeans
3 clusters
Silhouette 0.398

Machine Learning Pipeline

- 1

Data Augmentation
50 startups expanded to 1000 rows.
- 2

Preprocessing
StandardScaler + OneHotEncoder.
- 3

Regression
10 models benchmarked for Profit prediction.
- 4

Classification
Profit_Level: Low / Medium / High.
- 5

PCA
2D dimensionality reduction for visualization.
- 6

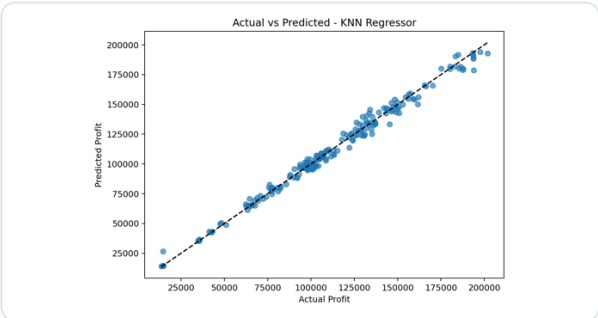
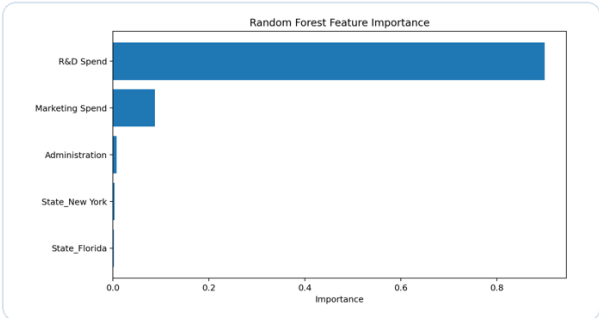
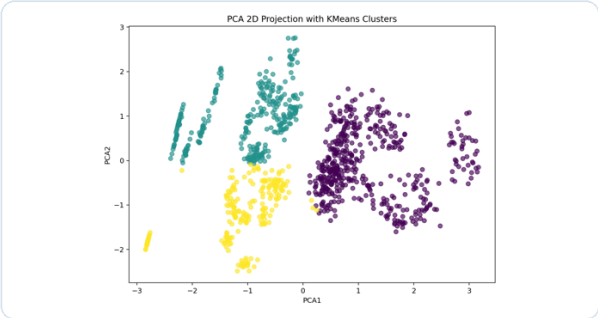
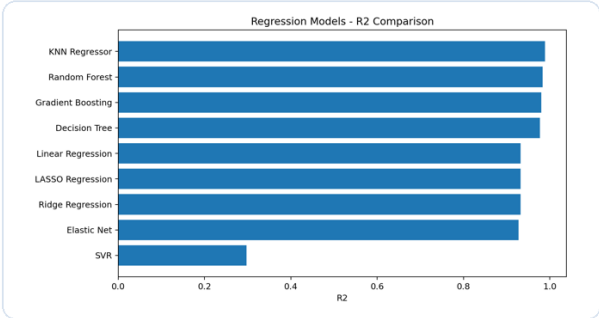
Clustering
KMeans finds startup groups.
- 7

Deployment
Best pipelines saved as joblib.

Top regression benchmark

Rank	Model	RMSE	R2
1	KNN Regressor	4,243	0.989
2	Random Forest	5,178	0.984
3	Gradient Boosting	5,732	0.980
4	Decision Tree	6,135	0.977
5	Linear Regression	10,545	0.932
6	LASSO Regression	10,545	0.932

Technical charts

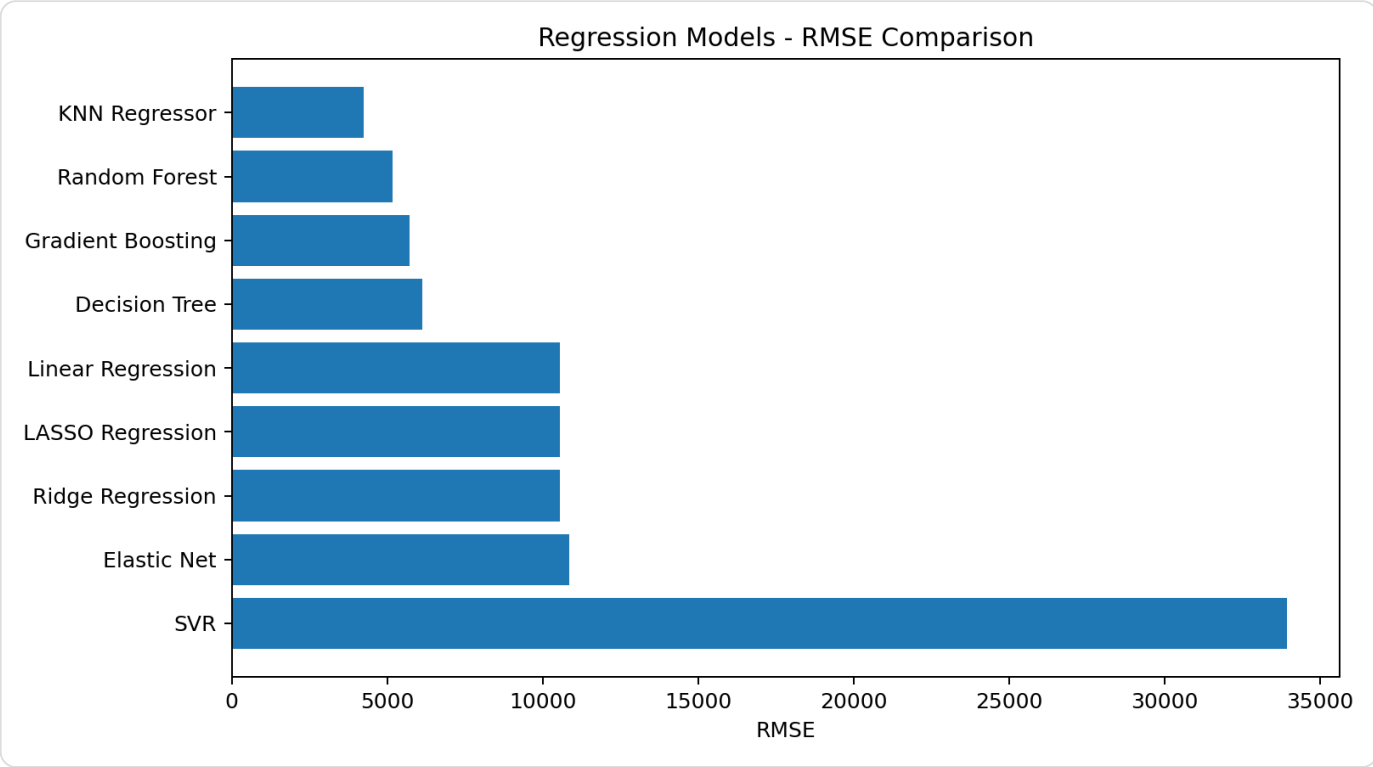
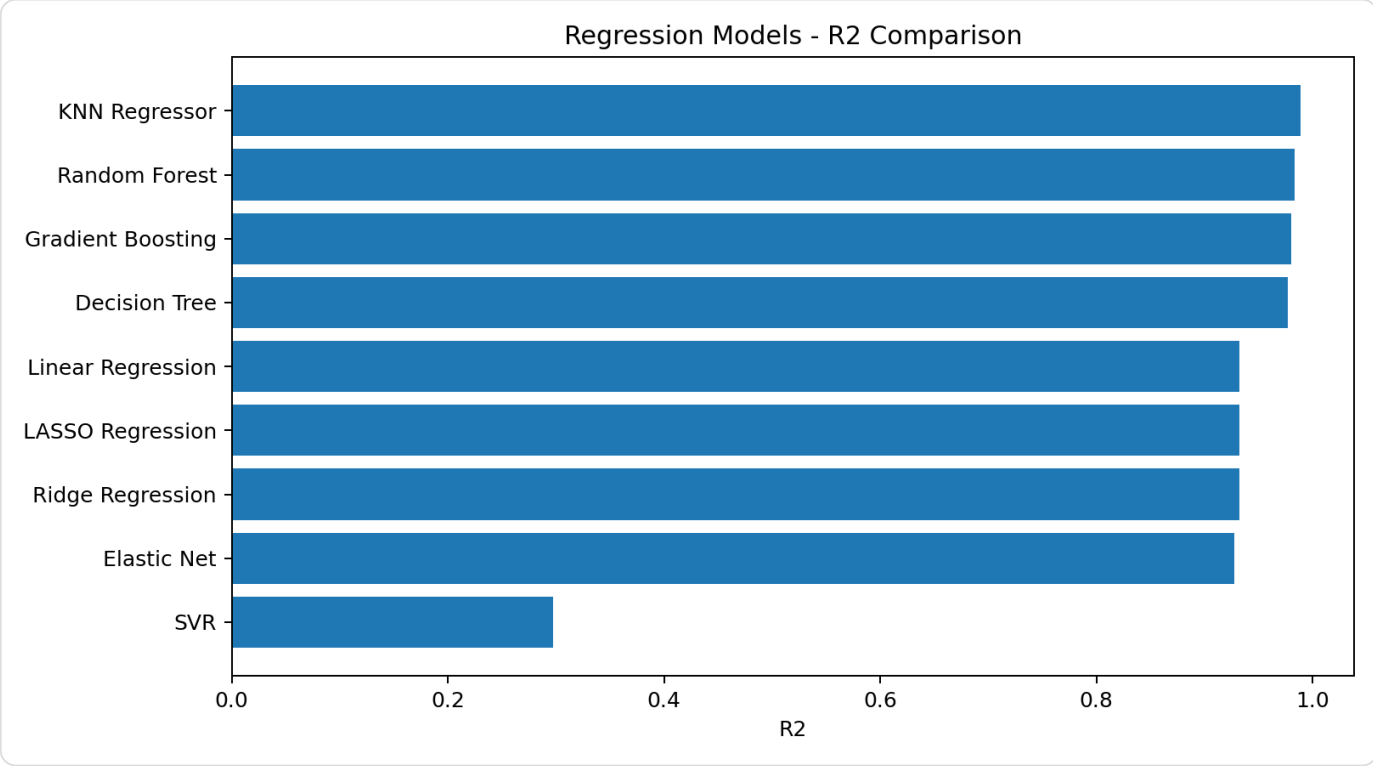


Final AI engineer conclusion

Best regression model is KNN Regressor. The project is ready for Streamlit or FastAPI deployment. Outputs include processed dataset, benchmark tables, charts, reports, and joblib pipelines.

Regression Benchmark

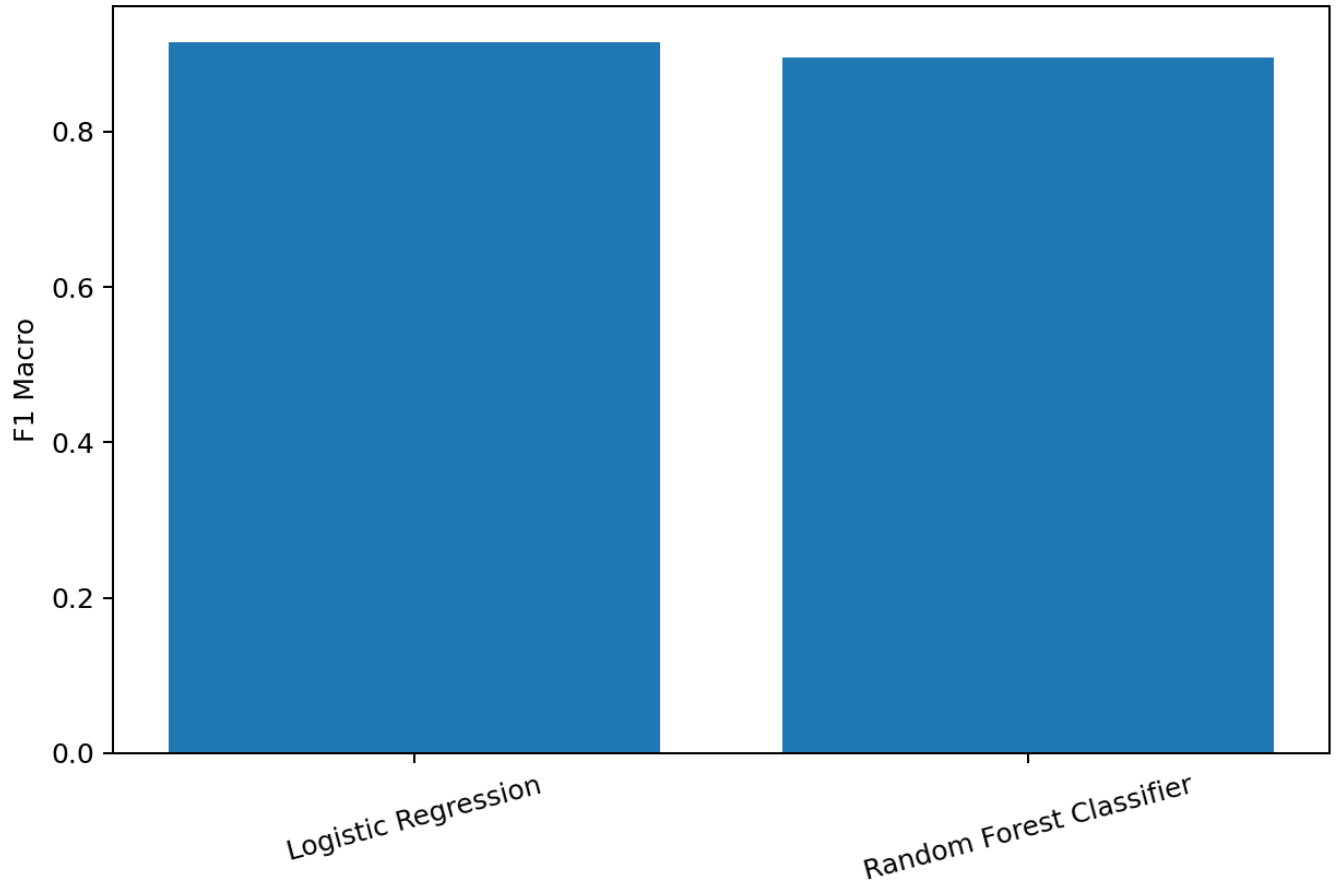
Model	MAE	RMSE	R2	CV_R2_Mean
KNN Regressor	3187.021845	4242.802407	0.989017	0.987992
Random Forest	3605.075876	5177.852987	0.983642	0.983399
Gradient Boosting	4191.041580	5732.364591	0.979951	0.978760
Decision Tree	4062.164954	6135.251138	0.977033	0.976985
Linear Regression	7737.051615	10545.402904	0.932149	0.935890
LASSO Regression	7737.046775	10545.406149	0.932149	0.935891
Ridge Regression	7739.055209	10548.293001	0.932112	0.935890
Elastic Net	7997.111042	10870.045526	0.927907	0.932798
SVR	24949.965225	33932.449963	0.297477	0.274968



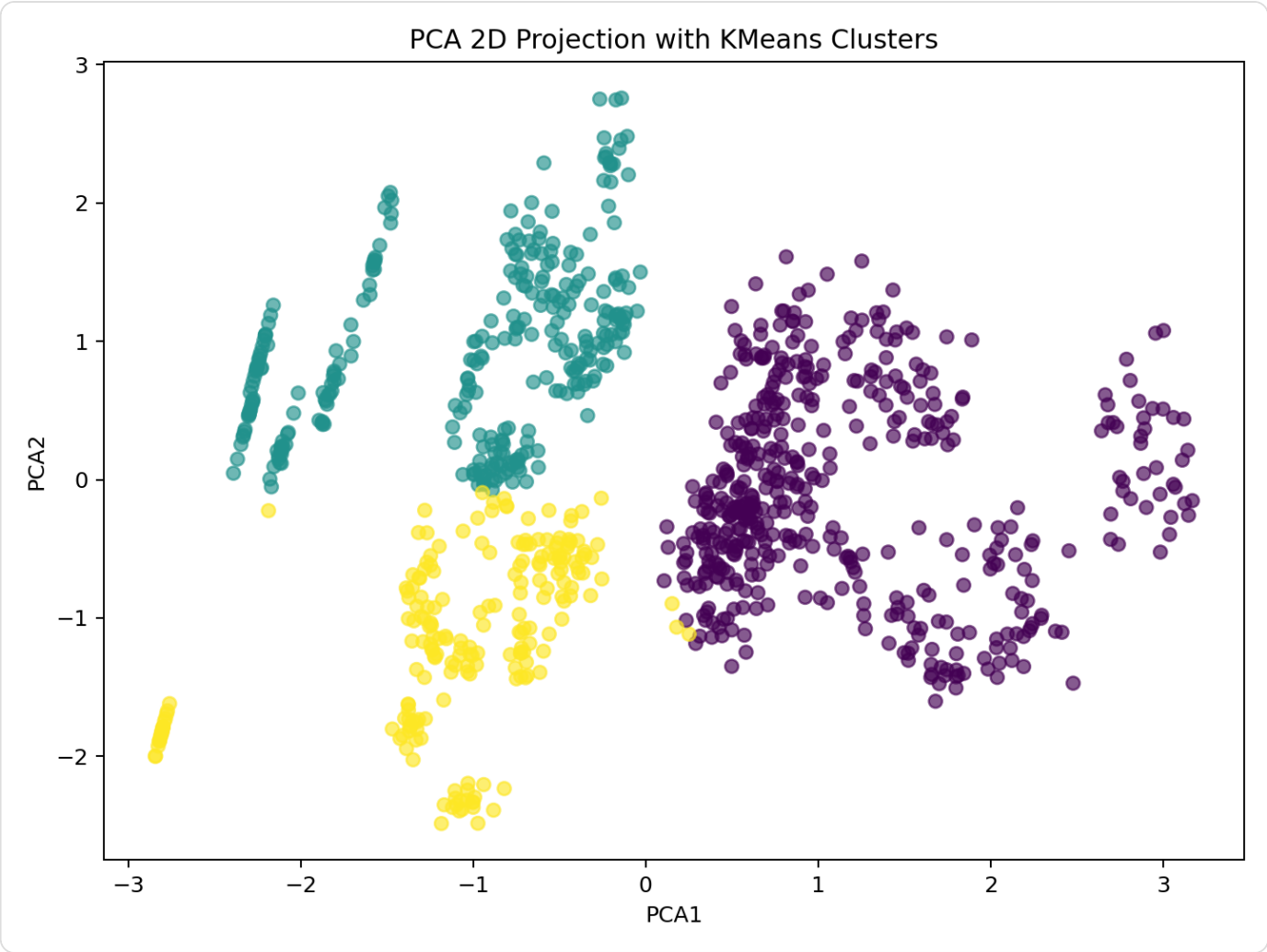
Classification Benchmark

Model	Accuracy	F1_Macro
Logistic Regression	0.915	0.915582
Random Forest Classifier	0.895	0.895799

Classification Models - F1 Macro

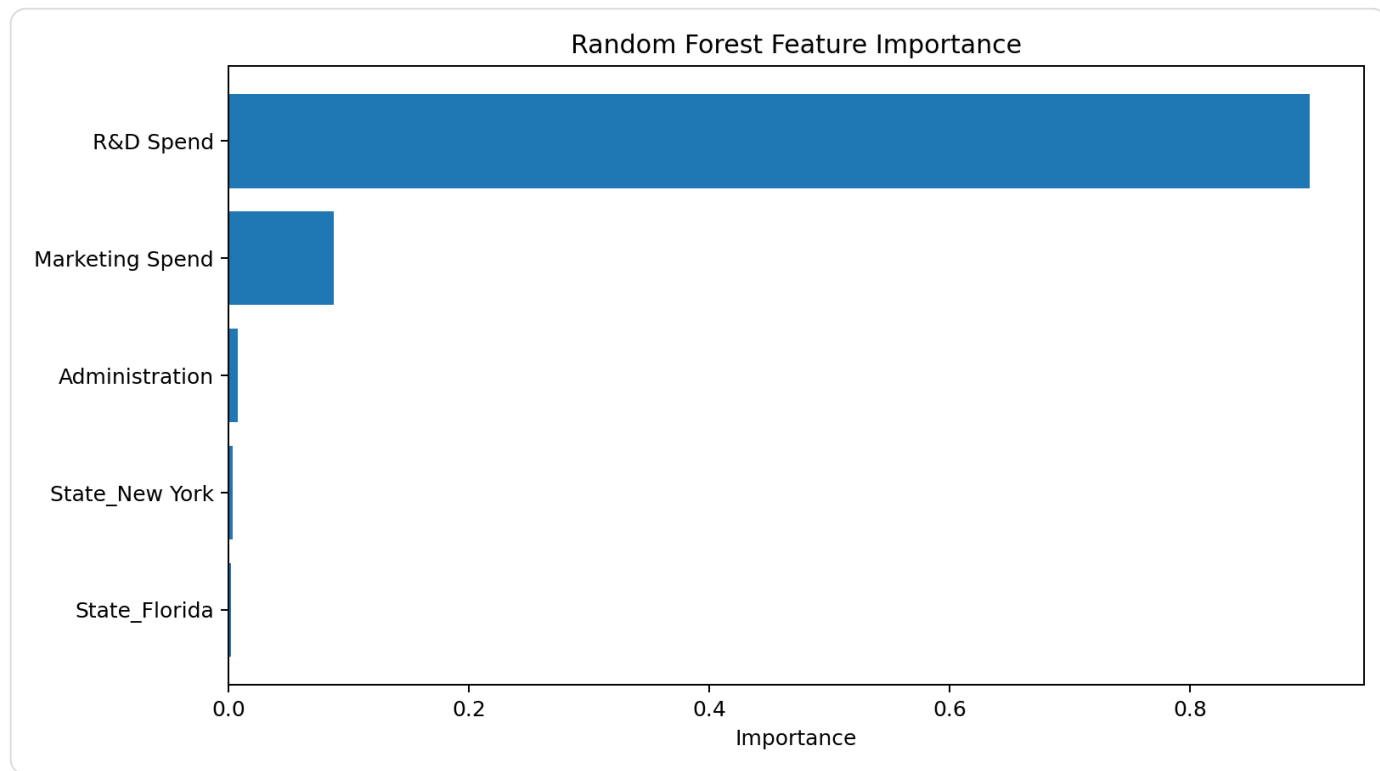


PCA and KMeans

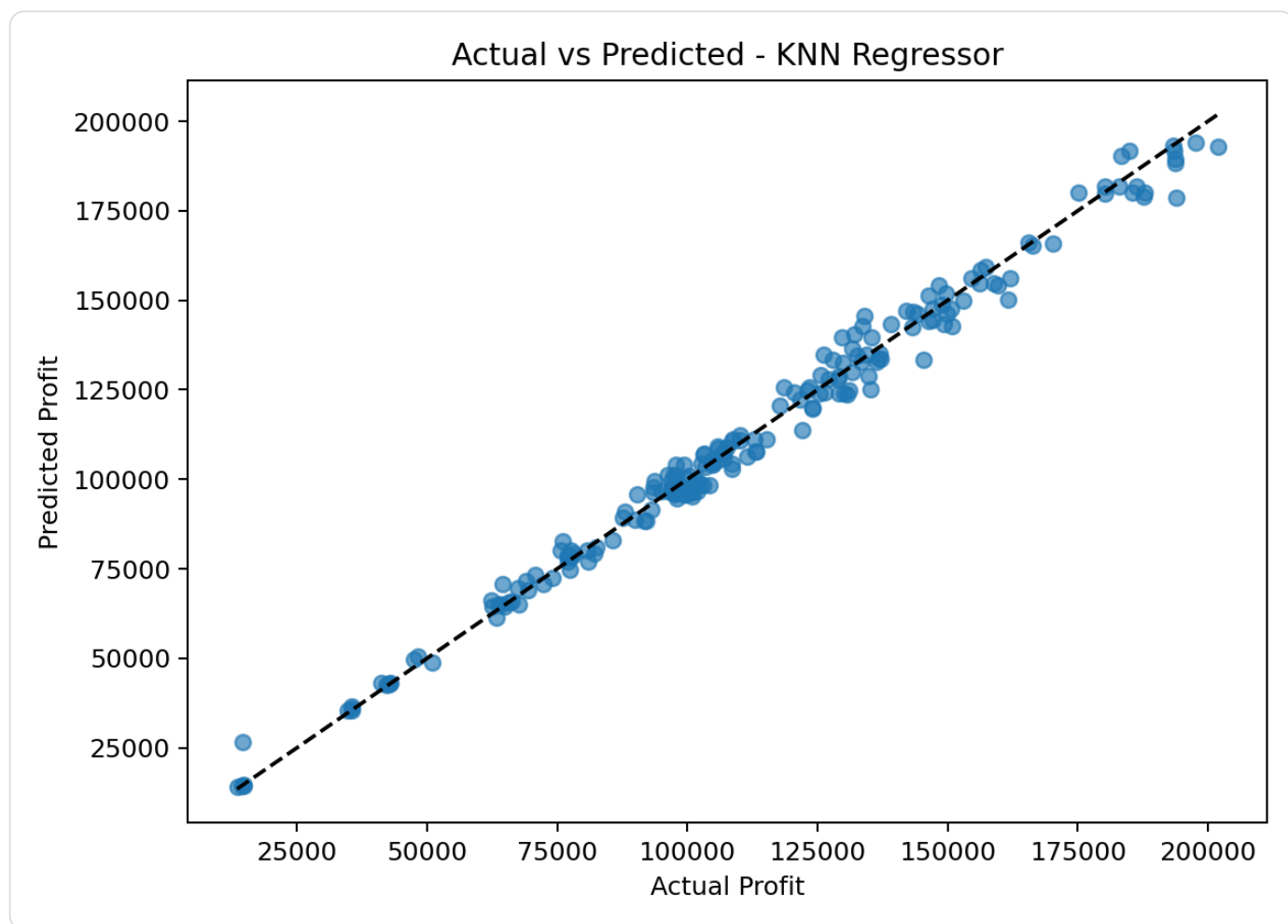


Feature Importance

Feature	Importance
R&D Spend	0.899785
Marketing Spend	0.087718
Administration	0.007385
State_New York	0.003305
State_Florida	0.001806



Actual vs Predicted



Markdown Report

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# HW6 NEW SKlearn - Full AI Engineer Project Report

## Executive Summary

This project upgrades the startup profit prediction task into a complete AI engineer-style machine learning project.

The same dataset is used for multiple machine learning tasks:

1. Data preprocessing
2. Regression prediction
3. Profit-level classification
4. PCA dimensionality reduction
5. KMeans clustering
6. Model benchmarking
7. Joblib deployment artifacts
8. Streamlit application

## Dataset

| Item | Value |
|---|---:|
| Rows | 1000 |
| Missing values | 0 |
| Numeric features | R&D Spend, Administration, Marketing Spend |
| Categorical feature | State |
| Regression target | Profit |
| Classification target | Profit_Level |

## Regression Results

| Model | MAE | RMSE | R2 | CV_R2_Mean |
|:-----:|-----:|-----:|-----:|-----:|
| KNN Regressor | 3187.02 | 4242.8 | 0.989017 | 0.987992 |
| Random Forest | 3605.08 | 5177.85 | 0.983642 | 0.983399 |
| Gradient Boosting | 4191.04 | 5732.36 | 0.979951 | 0.97876 |
| Decision Tree | 4062.16 | 6135.25 | 0.977033 | 0.976985 |
| Linear Regression | 7737.05 | 10545.4 | 0.932149 | 0.93589 |
| LASSO Regression | 7737.05 | 10545.4 | 0.932149 | 0.935891 |
| Ridge Regression | 7739.06 | 10548.3 | 0.932112 | 0.93589 |
| Elastic Net | 7997.11 | 10870 | 0.927907 | 0.932798 |
| SVR | 24950 | 33932.4 | 0.297477 | 0.274968 |

## Classification Results

| Model | Accuracy | F1_Macro |
|:-----:|-----:|-----:|
| Logistic Regression | 0.915 | 0.915582 |
| Random Forest Classifier | 0.895 | 0.895799 |

## PCA and Clustering

| Item | Value |
|---|---:|
| PCA components | 2 |
| Explained variance | 0.9231 |
| KMeans clusters | 3 |
| Silhouette score | 0.3976 |

## Feature Importance

| Feature | Importance |
|:-----:|-----:|
| R&D Spend | 0.899785 |
| Marketing Spend | 0.0877185 |
| Administration | 0.00738535 |
| State_New York | 0.0033054 |
| State_Florida | 0.00180577 |

## AI Engineering Conclusion
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The project is suitable as a complete sklearn homework and portfolio project. It demonstrates how a regression dataset can be extended into regression, classification, clustering, and dimensionality reduction tasks.

The final deployment design uses:

```
```text
Input startup spending data
 -> preprocessing pipeline
 -> best regression model
 -> predicted Profit
```
```

and

```
```text
Input startup spending data
 -> classifier pipeline
 -> Profit_Level
```
```